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Effect of intensified training on muscle ion kinetics, fatigue development, and repeated short-term performance in endurance-trained cyclists

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Gunnarsson TP, Christensen PM, Thomassen M, Nielsen LR, Bangsbo J. Effect of intensified training on muscle ion kinetics, fatigue development, and repeated short-term performance in endurance-trained cyclists. *Am J Physiol Regul Integr Comp Physiol* 305: R811–R821, 2013. First published July 24, 2013; doi:10.1152/ajpregu.00467.2012.—The effects of intensified training in combination with a reduced training volume on muscle ion kinetics, transporters, and work capacity were examined. Eight well-trained cyclists replaced their regular training with speed-endurance training (12 × 30 s sprints) 2–3 times per week and aerobic high-intensity training (4–5 × 3–4 min at 90–100% of maximal heart rate) 1–2 times per week for 7 wk and reduced training volume by 70% (intervention period; IP). The duration of an intense exhaustive cycling bout (EX2; 368 ± 6 W), performed 2.5 min after a 2-min intense cycle bout (EX1), was longer ($P < 0.05$) after than before IP (4:16 ± 0:34 vs. 3:37 ± 0:28 min:s), and mean and peak power during a repeated sprint test improved ($P < 0.05$) by 4% and 3%, respectively. Femoral venous K^+ concentration in recovery from EX1 and EX2 was lowered ($P < 0.05$) after compared with before IP, whereas muscle interstitial K^+ concentration and net muscle K^+ release during exercise was unaltered. No changes in muscle lactate and H^+ release during and after EX1 and EX2 were observed, but the in vivo buffer capacity was higher ($P < 0.05$) after IP. Expression of the ATP-sensitive K^+ (K_{ATP}) channel (Kir6.2) decreased by IP, with no change in the strong inward rectifying K^+ channel (Kir2.1), muscle Na^+-K^+ pump subunits, monocarboxylate transporters 1 and 4 (MCT1 and MCT4), and Na^+/H^+ exchanger 1 (NHE1). In conclusion, 7 wk of intensified training with a reduced training volume improved performance during repeated intense exercise, which was associated with a greater muscle reuptake of K^+ and muscle buffer capacity but not with the amount of muscle ion transporters.

potassium; lactate and proton release; interstitial potassium and high-intensity training

MUSCLE ION TRANSPORT PROTEINS affect the exchange of Na^+, K^+, H^+, Cl^-, and lactate across the sarcolemma during exercise and are of importance for fatigue development at different exercise intensities (8, 34, 36). Accumulation of extracellular K^+ causing sarcolemmal depolarization (41) and fatigue during intense exercise has been given a lot of focus in the past decades. During exercise, venous plasma K^+ concentration increases with increasing exercise intensity (21, 29, 45). During intense cycling and running, arm venous plasma K^+ levels up to ~9 mM have been observed (30, 45). During intense one-legged knee extensor exercise, interstitial K^+ concentrations exceeding 11 mM have been observed (33, 35). Exercise training can increase skeletal muscle Na^+-K^+ pump content in untrained subjects by 10–30% (28, 34) and reduce exercise-induced hyperkalaemia and interstitial K^+ accumulation (16, 34). However, in untrained individuals it is difficult to evaluate the cause of changes in protein expression and improvement in work capacity after a period of exercise training because it leads to adaptations in a significant number of physiological variables. Hence, new information may be obtained by changing the training in already trained subjects. Studies on well-trained subjects, who increased their training intensity, have reported increased Na^+-K^+ pump concentrations as determined by the [^{3}H]ouabain-binding technique (12, 27). Iaia et al. (18) and Bangsbo et al. (3) found that intensified training with a concomitant reduction in training volume improved short-term performance and increased the amount of the Na^+-K^+ pump subunits α_1 and α_2 leading to a lower arm venous K^+ concentration during intense running. However, it is unknown whether the training actually changes muscle K^+ kinetics. In addition, the impact of intense exercise training on the expression of human muscle K^+ channels, such as the strong inward rectifying K^+ channel (Kir2.1) and the ATP-sensitive K^+ (K_{ATP}) channel (Kir6.2), has not been investigated, and their role in human muscle K^+ homeostatis during exercise has not been examined.

Muscle ion transport proteins controlling lactate and H^+ efflux from the muscle cell may also be of importance for the working capacity of contracting muscle (4, 13), and a period of intensified training has been shown to increase the content of the monocarboxylate transporters (MCT1 and MCT4) and the Na^+/H^+ exchanger NHE1 in untrained subjects (20, 39). The adaptations in the transport proteins were associated with a lowered intracellular to interstitial gradient for lactate and H^+ (39) and a reduced extracellular to arterial lactate and H^+ gradient with a concomitant increase in lactate and H^+ release during intense exercise (20). Part of the increased lactate and H^+ release observed in the studies may be explained by an increase in the content of the membrane proteins, but other mechanisms are probably also involved. Hence, a part of the increase in lactate and H^+ release was due to an observed ~16% increase in thigh blood flow at exhaustion (38). The subjects in the latter studies were untrained, and it is unknown whether a period of intensified training in already trained subjects will affect lactate and H^+ release. In trained subjects intense training would not be expected to change blood flow (24, 46), thus any alteration in lactate and H^+ release may be mediated by changes in transport proteins involved in lactate and H^+ transport. Changes in MCT1 and NHE1 content in response to intensified training have been reported in some (6, 15, 18) but not all (3, 44) studies in trained runners and soccer players. Thus it is unclear whether intensified training in combination with a reduced training volume induces changes in muscle ion transport proteins involved in lactate and H^+ transport in trained athletes, and whether these changes leads to alterations in lactate and H^+ release during intense exercise. The in vivo buffer capacity is another factor of importance for

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change in muscle pH during exercise and has been shown to increase by training (40). However, changes in the in vivo buffer capacity in response to intensified training in combination with a reduced training volume has not been investigated in trained subjects. It may be hypothesized that the severe shifts in ion and lactate homeostasis observed during 30-s intervals (31) lead to an enhancement of the in vivo buffer capacity and thereby affect the work capacity during short-term intense exercise.

Thus the aims of the present study were to examine the effects of a period with intensified training in combination with a reduced training volume on muscle ion kinetics and work capacity and whether any changes relate to changes in muscle ion transport proteins. It was hypothesized that 7 wk of intensified training in combination with a reduced training volume induces changes in muscle ion transport proteins and thus alters intramuscular K^+ and H^+ handling during and after intense exercise.

METHODS

Subjects

Eight well-trained cyclists with an age, weight, and maximum oxygen uptake ($\dot{V}O_{2max}$) of 33.5 ± 2.9 yr, 81.0 ± 2.7 kg, and 59.1 ± 1.9 ml·kg^{-1}·min^{-1}, respectively, participated in the study. Subjects had been training and competing on a regular basis for at least 3 yr (range: 3–15). The study was approved by the Ethics Committee of Copenhagen and Frederiksberg communities and adheres to the principles of the Declaration of Helsinki and Title 45, US Code of Federal Regulations, Part 46, Protection of Human Subjects, Revised June 23, 2005, effective June 23, 2005. The subjects were informed of any risks and discomforts associated with the experiments before giving their written, informed consent to participate.

Experimental Design and Training

An intervention period (IP), with a one-group longitudinal design, lasting 7 wk was carried out immediately after the end of the cycling season (October to December). During a 4-wk (baseline) period before IP, subjects trained and competed 4–6 times/wk cycling 211 ± 29 km/wk. Training volume during IP was reduced by ~70% compared with the baseline period (62 vs. 211 km/wk). During IP subjects carried out 2–3 sessions of speed-endurance training [SET; 10–12 × ~30 s at 85–95% of peak power output (PPO) interspersed by 4.5 min recovery] and 1–2 sessions of aerobic high-intensity training [HI; 4–5 × 3–4 min at 90–95% of incremental PPO (iPPO) interspersed by 1.5–2 min of recovery, keeping a work-to-rest ratio of ~2:1] each week. All training sessions were performed on subjects' personal bikes. HI was performed on a 2-km straight course. Subjects were not allowed to draft during HI to keep up the highest possible power output during each interval. SET was performed uphill starting from a standing position and the recovery period between each exercise bout consisted of low-intensity (downhill) biking lasting ~4.5 min. SET sessions were always followed by a day of recovery. Power output (Powertap Rearwheel, CycleOps, Madison, WI) and heart rate (HR) of a SET and HI training session for a representative subject is shown in Fig. 1. All training sessions were supervised, HR was recorded during each session, and the subjects maintained their habitual lifestyle and recorded their training throughout IP as well as during the baseline period.

Four weeks before, and just before IP, subjects' endurance performance and $\dot{V}O_{2max}$ (see below) were determined to evaluate whether the performance of the subjects changed in the period leading up to IP. Before and after IP, subjects completed two performance tests separated by at least 48 h of recovery: *1*) an incremental test and *2*) a repeated sprint test (see *Testing Procedures*). In addition, subjects carried out two experimental days. On experimental days subjects completed exercise bouts at 50% and 70% of iPPO lasting 3–6 min, with resting periods of 30–60 min between bouts (Fig. 2). In addition, subjects completed two bouts at 90% of iPPO for 2 min (EX1) and to exhaustion (EX2), interspersed by 2.5 min of recovery (Rec 1). Immediately before each exercise bout subjects cycled for 2 min at 20 W. On the first experimental day femoral arterial and venous catheters were inserted into *a.* and *v. femoralis*, and muscle biopsies were taken at rest as well as before and immediately after EX1 and EX2. On the second experimental day, three microdialysis probes, six in total, were inserted into the *m. vastus lateralis* of each leg, and subjects completed 10 min of cycling at 20% of iPPO (82 ± 6 W) 30 min before the first exercise bout at 50% of iPPO. Furthermore, subjects completed two exercise bouts at 70% iPPO lasting 3 and 6 min and completed two bouts at 90% iPPO for 2 min and to exhaustion, interspersed by 2.5 min of recovery. Subjects rested for 30–60 min between each exercise bout (for further detail see *Experimental Procedures*).

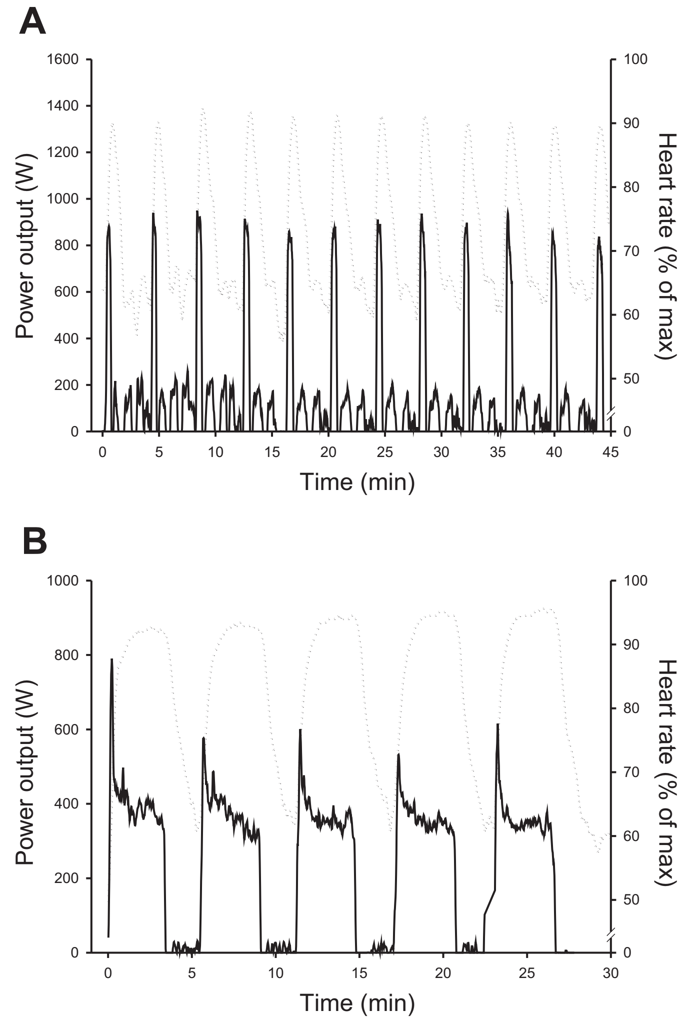

Fig. 1. Power output (W; solid line) and heart rate (expressed as % of maximal heart rate; dotted line) during a speed-endurance (*A*) and aerobic high-intensity (*B*) training session for a representative subject.

Testing Procedures

On the days of the performance tests subjects reported to the laboratory at least 3 h after consumption of a meal and testing was

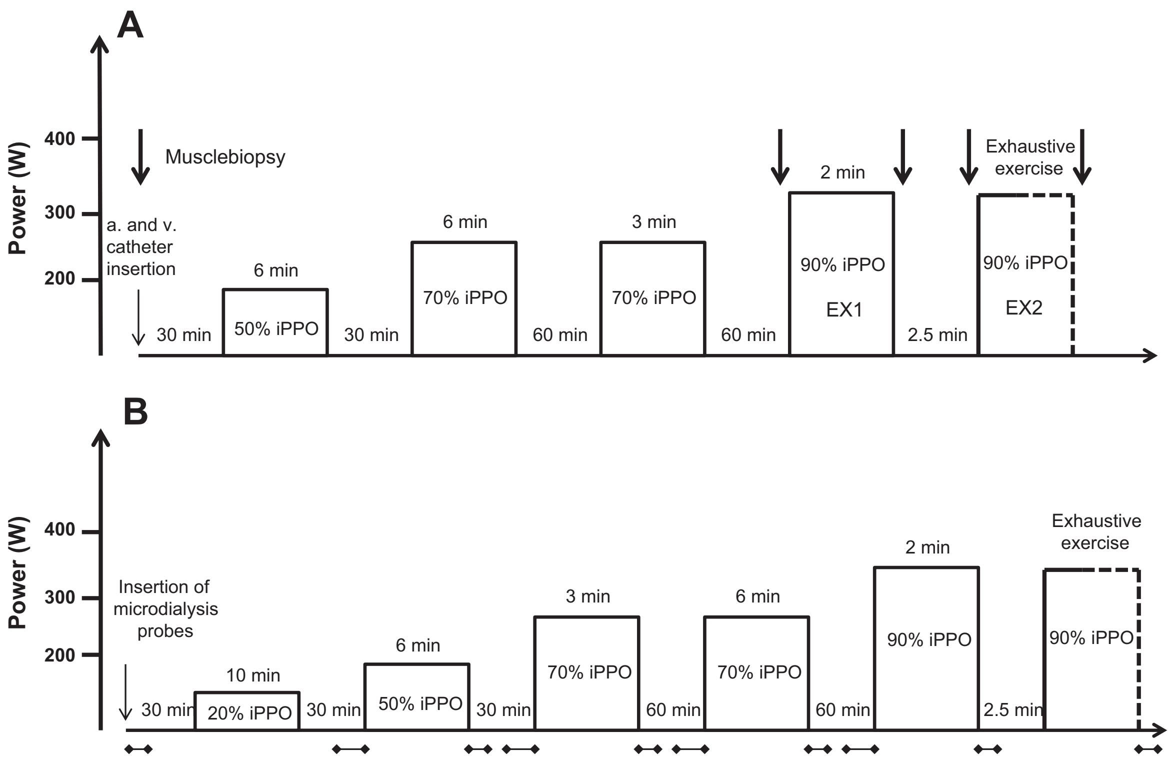

Fig. 2. Schematic presentation of experimental *days 1* (*A*; EXP1) and *2* (*B*; EXP2). iPPO, incremental peak power output.

completed indoor on an ergometer bike (Monark, Ergomedic 839E, Vansbro, Sweden).

Incremental test. Subjects cycled at 100 W for 6 min and, then work load was increased by 25 W/min until voluntary exhaustion defined as the time when the pedaling frequency dropped below 50 RPM. iPPO was calculated as the load (W) during the last completed work load + time completed on the last unfinished work load divided by 60 and multiplied by 25 W [e.g., iPPO for a subject performing 30 s at 400 W = 388 W (375 W + 30/60 × 25 W)]. At the incremental test before and after IP, subjects cycled for 10 min at 50% of iPPO (205 ± 3 W) followed by an increased workload of 25 W/min until voluntary exhaustion. Throughout the test pulmonary oxygen uptake was measured (Oxycon Pro, Viasys Healthcare, Hoechberg, Germany) and $\dot{V}O_{2max}$ was determined as the highest value achieved during a 30-s period. Criteria used for achievement of $\dot{V}O_{2max}$ were a plateau in $\dot{V}O_{2max}$ despite an increased load and a respiratory exchange ratio above 1.10.

Repeated sprint test. Subjects cycled for 6 min at 70% of iPPO (286 ± 6 W) where after they completed two 10-s sprints separated by 4 min of recovery. Ten minutes after the last sprint subjects completed 6 × 20-s all-out sprints separated by 100 s of recovery. The load on the bike was set to 0.75 N/kg and controlled by a computer. After each sprint subjects were instructed to continue pedaling and achieve a target cadence of ~80 rpm at 6 N (~50 W) within 20 s. This cadence was held constant until 10 s before the next sprint where the cadence was increased to 100 rpm. Five seconds before the start of each sprint the load was applied by the computer while subjects kept pedaling at 100 rpm. Subsequently, peak and mean power of the six sprints were calculated.

EXPERIMENTAL PROCEDURES

On the first experimental day (EXP1) subjects reported to the laboratory in the morning 60 min after consumption of a standardized breakfast. After 30 min of rest a catheter was inserted into the femoral vein under local anesthesia, using the Seldinger technique. The tip of the catheter was advanced to ~2 cm proximal to the inguinal ligament and a thermistor was advanced 8–10 cm proximal to the tip of the catheter (2) for measurements of blood flow. The catheter was used for blood sampling and to measure for blood flow by determining the drop in venous blood temperature. Another catheter was inserted into the femoral artery. Then a biopsy from the *m. vastus lateralis* was taken at rest. After 30 min of rest subjects cycled for 6 min at 50% of iPPO, then after 30 min of rest 6 min at 70% of iPPO and 60 min later 3 min at 70% of iPPO. Then, after another 60 min of rest, subjects performed two exercise bouts at an intensity corresponding to 90% of iPPO (368 ± 6 W) interspersed by 2.5 min of recovery. The first exercise bout (EX1) lasted 2 min and the second bout (EX2) was performed to exhaustion. During the 2.5-min recovery period the subject were sitting on the bike. Before the exercise bouts subjects cycled for 2 min at 20 W. A muscle biopsy was taken at rest and before and immediately after EX1 and EX2. Arterial and venous blood samples were drawn at rest, 10 s before EX1, and at 0.17, 0.50, 1.00, 1.50, and 1.83 min during and at 0.17, 0.50, 1.00, 1.50, 1.83, and 2.33 min in recovery from EX1, respectively. Samples were drawn 10 s before EX2 and at 0.17, 0.50, 1.00, 1.50, and 1.83 min samples, with continuous sampling every 20 s until exhaustion as well as 0.17, 0.50, 1.00, 1.50, 2.00, 3.00, and 5.00 min after EX2, respectively.

On the second experimental day (EXP2) subjects reported to the laboratory in the morning after a standardized breakfast. After 30 min of rest in the supine position local anesthesia (1 ml of lidocaine, 20 mg/ml without epinephrine) was administered to the skin and subcutaneous tissue. Three microdialysis probes were inserted in the *m. vastus lateralis* of each leg in a direction parallel with the muscle fiber orientation (17). Membrane length, outer diameter, and molecular cut-off of the probes (custom made) were 40 mm, 0.22 mm, and 6,000 Da, respectively. The positioned probes were photographed, and the distance from patella and the anterior superior iliac spine was measured. Probes were placed in similar positions during posttesting. After insertion, the microdialysis probes were perfused at a rate of 5 μl/min for 1 h using high precision pumps (Harvard and World

Precision Instruments) with an isotonic Ringer acetate solution (Pharmacia, Stockholm, Sweden). Thirty minutes after insertion of the probes subjects cycled for 10 min at 20% of iPPO (82 ± 6 W). After an additional 30 min of rest while in the supine position, subjects cycled for 6 min at 50% of iPPO and after 30 min of rest 3 min at 70% of iPPO. After resting for another 60 min, subjects cycled for 6 min at 70% of iPPO. After 60 min rest, subjects performed two intense exercise bouts at 90% of iPPO. During the first bout subjects cycled for 2 min (EX1) and after 2.5 min of rest to exhaustion during the second bout (EX2). Before the EX1 and EX2, subjects cycled for 2 min at 20 W, and during the 2.5 min of recovery between EX1 and EX2 subjects remained on the bike. Microdialysis sample tubes were weighed before and after sampling to determine the actual flow, and only probes with a calculated perfusion rate of 3.5–5.5 μl/min were used. Microdialysis dialysate was collected before and immediately after EX1 in 45-s intervals over 2.25 min of time. In recovery from EX2 six samples were collected over 6.5 min of time. During the first 1.5 min, two samples were collected (45 s per sample): during the following 2 min two samples were collected (1 min per sample); and during the last 3 min two samples were collected (1.5 min per sample). After being weighed, the samples were stored at −20°C until analysis. On all experimental days pulmonary $\dot{V}O_2$ was measured breath by breath (Oxycon Pro, Viasys Healthcare, Hoechberg, Germany).

Blood Analysis

Immediately after being sampled, a part of the blood was centrifuged at 20,000 *g* for 30 s, transferred to Eppendorf tubes, and then placed in ice-cold water until being stored at −20°C. Samples were subsequently analyzed for K^+ by an ion-selective electrode using a Hitachi 912 Automatic Analyzer (Roche Diagnostic). The rest of the blood sample was immediately analyzed for blood lactate, H^+, hemoglobin, and hematocrit (ABL 800 Flex, Radiometer, Copenhagen, Denmark).

Microdialysate

Microdialysate samples were analyzed for K^+ on a flame photometer (FLM3, Radiometer) using lithium as an internal standard.

Muscle Analysis

The muscle samples were immediately frozen in liquid N_2 and stored at −80°C. The frozen muscle tissue samples were weighed before and after freeze drying to determine the water content. After freeze drying was completed, connective tissue, visible fat, and blood were carefully dissected away. Dissecting was done under a stereo microscope with an ambient temperature of ~18°C and a relative humidity below 30%.

Muscle ion transport proteins. A part of the muscle sample taken at rest (~4–5 mg dry wt) was homogenized on ice in a fresh batch of buffer (10% glycerol, 20 mM Na-pyrophosphate, 150 mM NaCl, 50 mM HEPES, 1% Nonidet P-40, 20 mM β-glycerophosphate, 10 mM NaF, 2 mM PMSF, 1 mM each of EDTA and EGTA, and 10 μg/ml each aprotinin and leupeptin and 3 mM benzamidine) with a Polytron 3100 (Kinematica) for not more than 30 s. After rotation end over end for 1 h, the samples were centrifuged for 30 min at 17,500 *g* at 4°C, and lysates were collected as the supernatant. Protein concentrations were determined in lysates by ELISA using BSA standards (Pierce Reagents). The lysates were diluted to appropriate protein concentrations in a 6× sample buffer (0.5 M Tris-base, DTT, SDS, glycerol, and bromphenol blue), and equal amounts of total protein (5–15 μg in accordance with the antibody optimization) were loaded for each sample in different wells on 10% or 16.5% precasted Tris·HCl gels (Bio-Rad Laboratories). For comparisons, samples from the same subject were always loaded on the same gel. Loading controls were not used because *1*) we have previously observed that loading controls such as actin and cav3 are changing with training interventions similar to the one used in the present study (unpublished data; Thomassen M and Gunnarsson TP), *2*) protein concentrations were determined in lysates (see above), and *3*) as at least two proteins are analyzed per gel a loading artifact would be displayed. The gel electrophoresis ran for 80–100 min at 55 mA and a maximum of 150 V per gel. Afterwards proteins were blotted to a polyvinylidene difluoride membrane at 70 mA and a maximum of 25 V per gel in ~2 h. The membranes were incubated overnight with 15–30 ml of primary antibody diluted in either 2% nonfat milk [monoclonal Na^+-K^+ pump α_1-subunit (~100 kDa), 1:500 dilution (C464.6, no. 05-369, Merck Millipore); polyclonal α_2-subunit (~100 kDa), 1:500 dilution (no. 07-674, Merck Millipore), monoclonal β_1-subunit (~50 kDa), 1:1,000 dilution (MA3–930, Affinity BioReagents) and polyclonal K_{ATP} channel (Kir6.2) (~50 kDa), 1:500 dilution (Sc-11228, Santa Cruz Biotechnology)] or 3% BSA [monoclonal NHE1 (~100 kDa), 1:500 dilution; polyclonal MCT1 (~43 kDa), 1:1,000 dilution, polyclonal MCT4 (~43 kDa), 1:1,000 dilution (MAB3140, AB3538P, and AB3316P, Merck Millipore) and polyclonal inward rectifier K^+ channel (Kir2.1) (~50 kDa), 1:1,000 dilution, (AB5374, Merck Millipore)]. Samples used for determination of Kir2.1 were heated to 96°C for 2 h, whereas

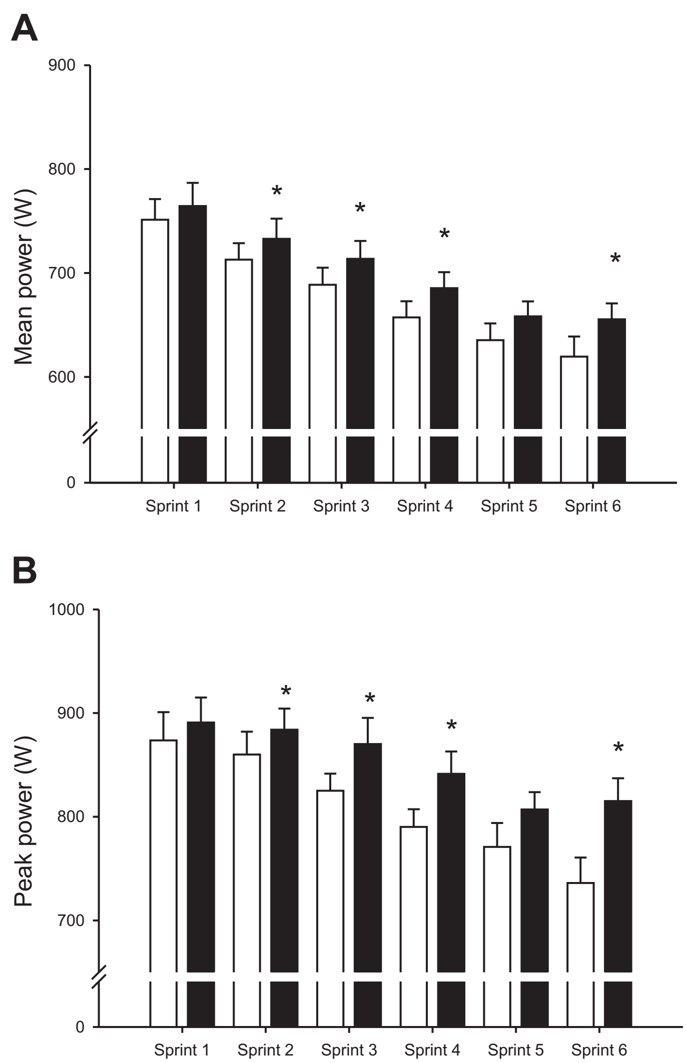

Fig. 3. Mean (*A*) and peak (*B*) power output (W) during 6 repeated 20-s sprints interspersed by 100 s of recovery before (open bars) and after (filled bars) 7 wk of speed-endurance training in 8 trained cyclists. Values are means ± SE. *Different ($P < 0.05$) from before.

all other samples were nonheated before loading. After being washed briefly in a Tris-buffered saline (Tween), membranes were incubated with a secondary antibody for ~1 h at room temperature. The secondary horseradish peroxidase-conjugated antibodies used were diluted 1:5,000 in 2% nonfat milk or 3% BSA depending on the primary antibody (P-0447, P-0448, and P-0449, DakoCytomation). The membrane staining was visualized by incubation with a chemiluminescent horseradish peroxidase substrate (Merck Millipore) immediately before the image was digitalized on a Chemi Doc MP (Bio-Rad Laboratories, Hercules, CA). Net band intensities were quantified using Image Lab (Image Lab v. 4.0, Bio-Rad Laboratories). Changes in protein expression were determined from double determinations, i.e., the biopsies were divided and kept in two parts before freeze drying, resulting in two results for the same time point. The mean signal intensity of the two samples was used as the result for the individual time point. The intensity of the individual time points were divided with the mean intensity of the prevalues within the group to show the variation in the prebiopsies, and data are expressed as geometric means ± 95% confidence intervals.

Intramuscular ions. A part of the muscle sample (1–2 mg dry wt) obtained on EXP1, immediately before and after EX1 and EX2, were wet washed with H_2O_2 (Merck) at a temperature of 90°C and then dissolved in 400 μl 0.6 M PCA and 600 μl 5 mM LiCl. Intramuscular K^+ content was determined by flame photometry (FLM3, Radiometer).

Muscle creatine phosphate, lactate, and pH. A part of the muscle sample (~1 mg dry wt), obtained on EXP1 immediately before end after EX1 and EX2, was homogenized in a buffer containing 0.6 M perchloric acid (PCA) and 1 mM EDTA, neutralized to pH 7.0 with 2.2 M $KHCO_3$, and stored at −80°C until analyzed for muscle lactate and creatine phosphate (CP) fluorometrically (25). A part of the muscle sample (~1 mg of dry wt) obtained on EXP1 immediately before and after EX1 and EX2 was homogenized in a 120 μl nonbuffering solution containing 145 mM KCl, 10 mM NaCl, and 5 mM iodoacetic acid, and muscle pH was measured using a small glass electrode (Radiometer GK2801).

Calculations

Net plasma K^+ from the thigh was calculated as LBF × $\{[K^+]_v \times (1 - Hct_v) - [K^+]_a \times (1 - Hct_a)\}$, where LBF is the leg femoral venous blood flow (l/min) and $[K^+]$ is the concentration of K^+. Because of variation in determination of LBF and since LBF was not different before and after IP, the average LBF was used in the

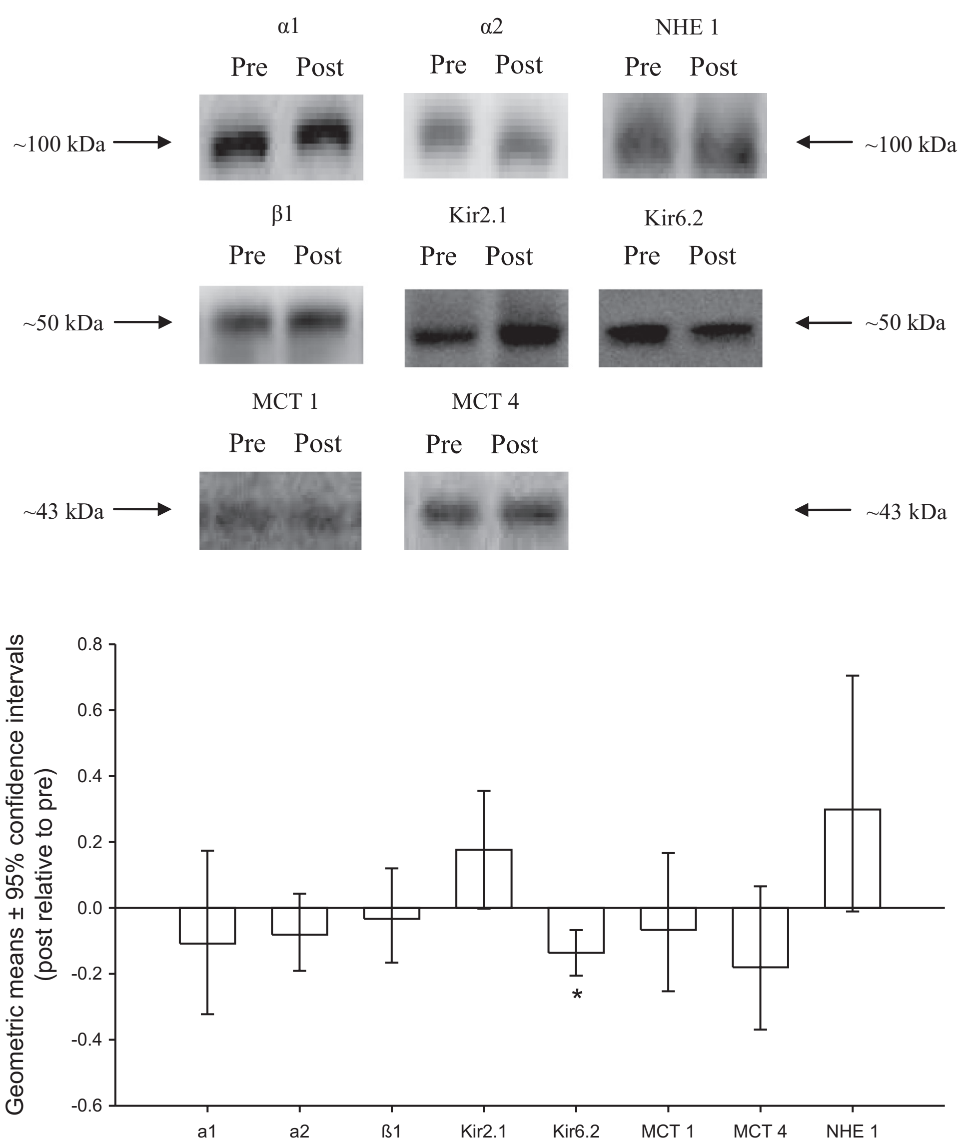

Fig. 4. Muscle Na^+-K^+ pump α_1, α_2, and β_1 subunits, inward rectifying K^+ channels (Kir2.1 and Kir6.2), monocarboxylate transporters (MCT) 1 and 4 and Na^+/H^+ exchanger (NHE) 1 before and after 7 wk of high-intensity training in combination with a reduced training volume in trained cyclists (n = 8). Representative blots at ~43, ~50, and ~100 kDa are shown. Data are expressed as geometric means (after relative to before) ± 95% confidence intervals.

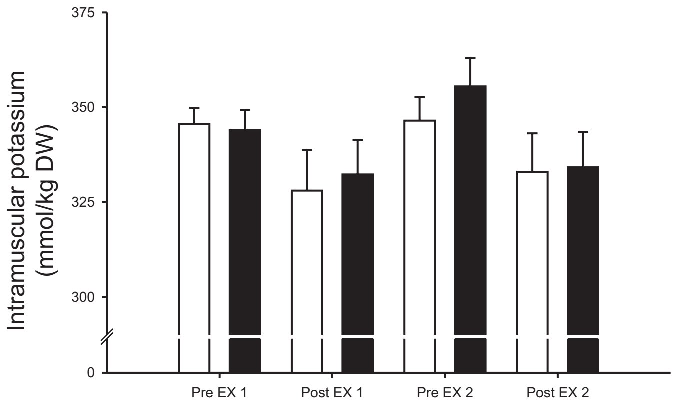

Fig. 5. Intramuscular potassium concentrations before (Pre) and after (Post) cycling at 20 W followed by EX1 and EX 2 (368 ± 6 W) before (open bars) and after (filled bars) 7 wk of high-intensity training in combination with a reduced training volume in trained cyclists (n = 7). Values are means ± SE.

Table 1. *Net potassium release (mmol) during EX1 and the first 2 min of EX2 as well as the first 2 min of recovery from EX1 (Rec 1) and EX 2 (Rec 2) before and after 7 wk of high-intensity training in combination with a reduced training volume in trained cyclists*

	Before	After
EX1	2.7 ± 0.8	2.3 ± 1.0
Rec 1	−1.0 ± 0.7	−2.0 ± 0.9
EX2	4.5 ± 1.0	1.7 ± 1.6
Rec 2	−0.6 ± 0.4	−1.6 ± 0.6

Values are means ± SE; n = 7 cyclists. EX1 and EX2, first and second exercise bout, respectively; Rec 1 and Rec 2, recovery 1 and 2, respectively.

calculation. The total net release of K^+ during EX1 and the first 2 min of EX2 as well as the first 2 min of recovery from EX1 and EX2 were calculated as the area under the curve. Net blood lactate release from the thigh was calculated by multiplying the difference in concentrations of lactate between the femoral artery and vein (a-v diff) with LBF. When calculating the time of the microdialysate, it was taken into account that there was an estimated 30-s delay between the outlet of the machine and point of sample collection (34). Sampling after EX1 (0–45, 45–90, and 90–135 s) represents muscular sampling of −7.5, 37.5, and 82.5 s, respectively (as there is a 30-s delay and each sample averages). The immediate sample averages 22.5 s (45 s sampling) and with a 30-s delay represents a muscular time point of −7.5 s (7.5 s before end exercise).

The intracellular K^+ concentration (mM) was calculated as $[K_i^+] = [K_m^+]/(1 - w_e) - [K_{in}^+] \times w_e$, where $[K_m^+]$ is the intramuscular K^+ concentration, w_e is extracellular water content that is assumed to be 12% before EX1 and 15% after EX1, as well as before and after EX2 (43), and $[K_{in}^+]$ is the interstitial K^+ concentration (mM). The net H^+ release of the leg was calculated as $(SBE_v - SBE_a) \times$ LBF, where arterial (a) and venous (v) standard base excess (SBE) were calculated as SBE $= (1 - [Hb] \times 0.023) \times ([HCO_3^-] - 24.4) + (2.3 \times [Hb]) + (7.7) \times ([pH] - 7.4)$, where [Hb] is the hemoglobin concentration, $[HCO_3^-]$ is the plasma bicarbonate concentration (42), and [pH] is the pH value.

Statistics

Performance during the repeated sprint test was evaluated using a two-way analysis of variance (ANOVA) for repeated measures with time (before vs. after) and sprints (numbers 1–6) as factors. Intramuscular and interstitial K^+ and K^+, lactate, and H^+ release were evaluated using a two-way ANOVA for repeated measures with time (before vs. after) and sampling time as factors. When a significant interaction was detected, data were subsequently analyzed using a Student-Newman-Keuls post hoc test. Changes in muscle ion transport proteins, arterial and venous plasma K^+ and net K^+, lactate, and H^+ release at rest and before, during, and after EX1 and EX2 as well as the in vivo muscle buffer capacity during EX1 and EX2 were evaluated using a Student's paired t-test (before vs. after).

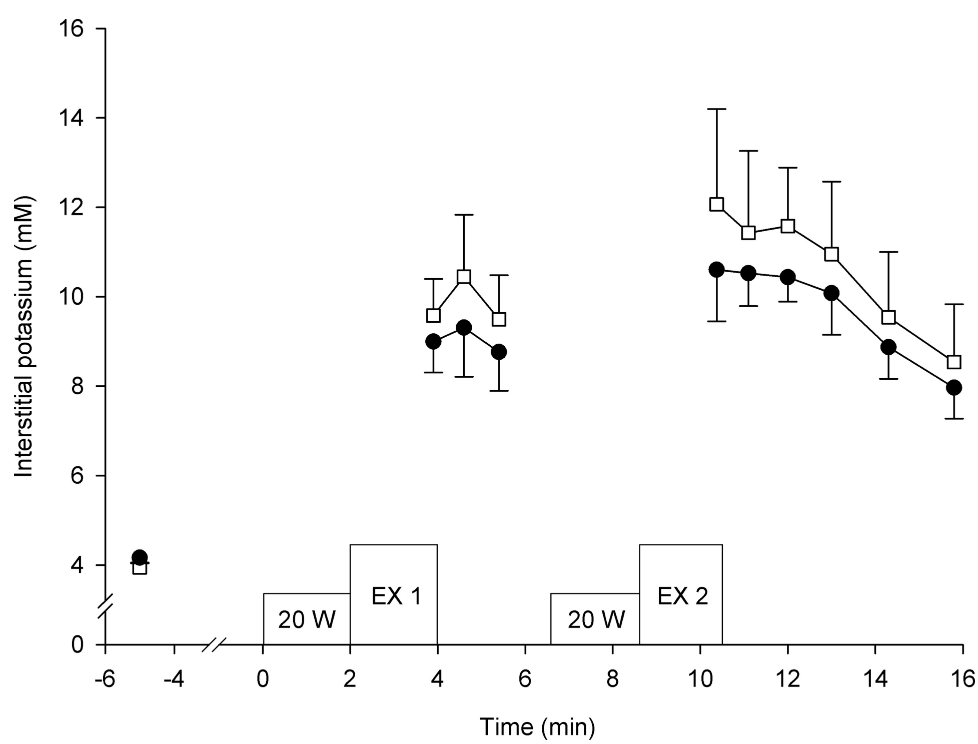

Fig. 6. Interstitial potassium concentrations before and after cycling at 20 W followed by EX1 and EX2 (368 ± 6 W) before (open symbols) and after (filled symbols) 7 wk of high-intensity training in combination with a reduced training volume in trained cyclists (n = 7). Values are means ± SE.

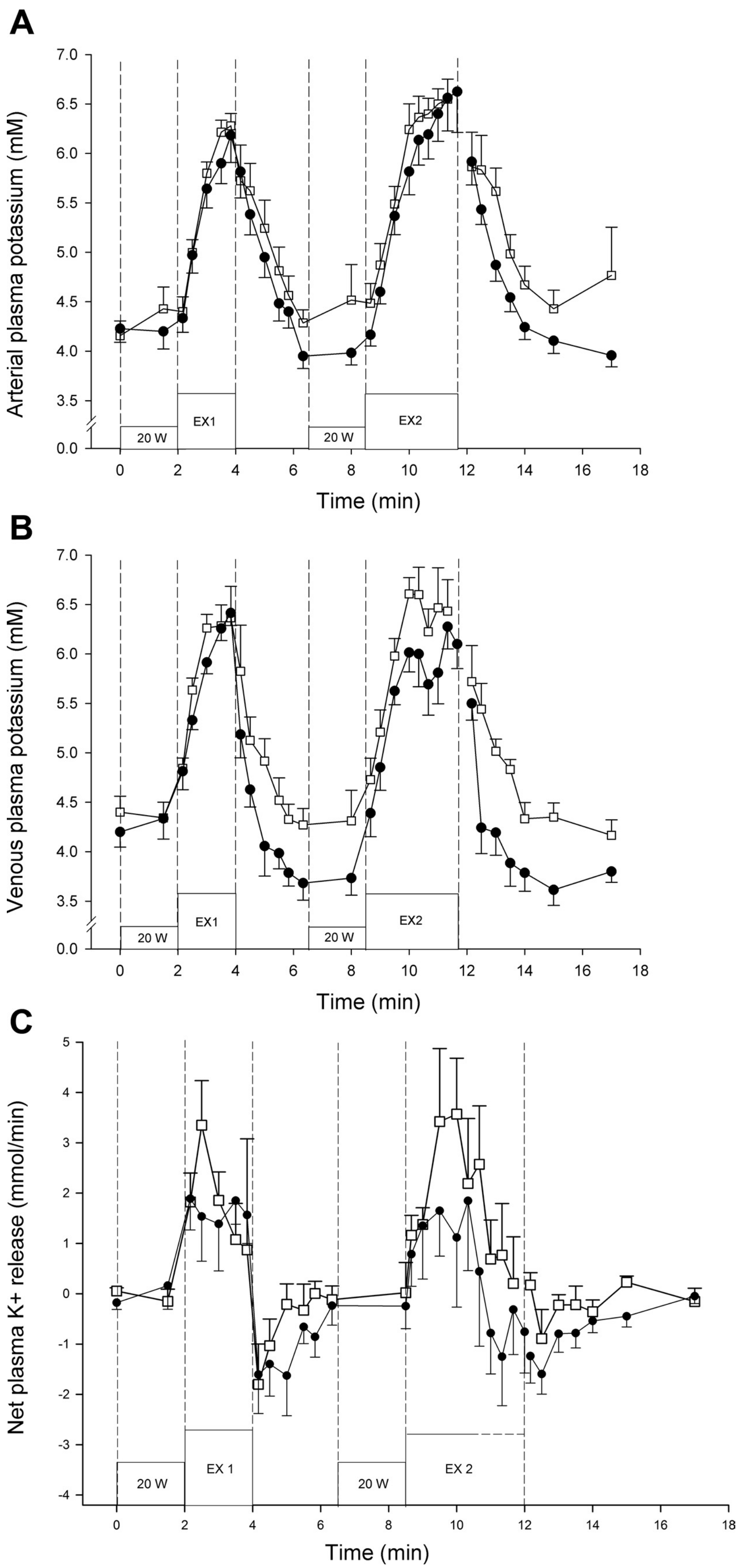

Fig. 7. Arterial (*A*) and venous (*B*) plasma potassium and net plasma potassium release (*C*) before, during, and after cycling at 20 W followed by EX1 and EX2 (368 ± 6 W) before (open symbols) and after (filled symbols) 7 wk of high-intensity training in combination with a reduced training volume in trained cyclists (n = 7). Values are means ± SE.

Table 2. *Average arterial and venous plasma potassium (mM) during the first 2 min of EX1 and EX2 as well as the first 2 min of recovery from EX1 (Rec 1) and EX2 (Rec 2) before and after 7 wk of high-intensity training in combination with a reduced training volume in trained cyclists*

	Before	After
EX1		
Arterial	5.5 ± 0.1	5.3 ± 0.2
Venous	5.9 ± 0.1	5.7 ± 0.1
Rec 1		
Arterial	5.2 ± 0.3	4.9 ± 0.2
Venous	4.9 ± 0.2	4.2 ± 0.2*
EX2		
Arterial	5.5 ± 0.2	5.2 ± 0.2
Venous	5.8 ± 0.1	5.4 ± 0.2
Rec 2		
Arterial	5.3 ± 0.2	4.8 ± 0.1
Venous	5.1 ± 0.1	4.3 ± 0.2*

Values are means ± SE; $n = 7$ cyclists. *After different ($P < 0.05$) from before.

RESULTS

Exercise Capacity

Time to exhaustion during EX2 was longer ($P < 0.05$) after compared with before IP (256 ± 34 vs. 217 ± 28 s). After IP, average mean and peak power output during the repeated sprint test was elevated ($P < 0.05$) by 4% and 3%, respectively, and mean and peak power output during sprint 2, 3, 4, and 6 were higher ($P < 0.05$) after compared with before IP (Fig. 3).

Muscular Adaptations

After IP, the content of Kir6.2 was 14% lower ($P < 0.01$) than before IP, whereas no significant changes in Kir2.1 (18%, $P = 0.06$), NHE1 (30%, $P = 0.09$), and the Na^+-K^+ pump subunits α_1, α_2, and β_1 or in MCT1 and MCT4 were observed by IP (Fig. 4).

K^+ Response to Exercise

No differences in intramuscular K^+ concentration before and after IP were observed, and the decrease in intramuscular K^+ during exercise was similar before compared with after IP during both EX1 (18 ± 6 vs. 12 ± 4 mmol/kg dry wt) and EX2 (12 ± 8 vs. 22 ± 9 mmol/kg dry wt) (Fig. 5). The intracellular K^+ concentration before EX1 was 125 ± 5 and 129 ± 3 mM before and after IP, respectively, and 110 ± 19 and 114 ± 5 mM after EX1. Before EX2 the intracellular K^+ concentration was 136 ± 6 and 118 ± 4 mM before and after IP, respectively, and 114 ± 4 and 110 ± 5 mM after EX2, respectively. The interstitial K^+ concentration before, at the end of, and after EX1 and EX2 did not change with IP, reaching 11–12 mM at the end of EX2 (Fig. 6).

The average venous K^+ concentration during the first 2 min of recovery from EX1 and EX2 was lower ($P < 0.05$) after compared with before IP (Table 1; Fig. 7). Net K^+ release was <0.3 mmol/min at rest and increased within the first 30 s of EX1 to 3.3 ± 0.9 and 1.9 ± 0.6 mmol/min before and after IP, respectively, and was 0.9 ± 2.2 and 1.6 ± 0.6 mmol/min at the end of EX1 (Fig. 7). No significant difference in total net K^+ release during EX1 (2.3 ± 1.0 vs. 2.7 ± 0.8 mmol) and the first 2 min of EX2 (4.5 ± 1.0 vs. 1.7 ± 1.6 mmol) was observed between before and after IP (Table 2).

Lactate Response to Exercise

Muscle lactate before EX1 and EX2 was the same after compared with before IP, but both during EX1 and EX2 muscle lactate increased more ($P < 0.05$) after compared with before IP. Furthermore, the rate of muscle lactate clearance in recovery from EX1 was higher ($P < 0.05$) after compared with before IP (Table 3). No difference in arterial and venous lactate was observed during or in recovery from EX1 and EX2. During EX1 peak arterial lactate was 16.8 ± 1.5 and 15.1 ± 1.9 mM and peak venous lactate was 15.6 ± 1.6 and 13.2 ± 0.3 mM after and before IP, respectively (Fig. 8). Net lactate release during and in recovery from EX1 and EX2 was the same before and after IP (Fig. 8).

H^+ Response to Exercise

Muscle pH before and after EX1 and EX2 was not different before and after IP. Estimated net H^+ release at the end of EX1 was 34.5 ± 6.7 and 23.2 ± 8.1 mmol/min before and after IP, respectively, and it was −3.4 ± 5.2 and 2.6 ± 2.5 before EX2 and peaked at 47.8 ± 25.2 and 40.8 ± 7.1 mmol/min during EX2 (Fig. 9). The ratio between H^+ and lactate release at the end of EX1 was 3.1 ± 2.0 and 13.0 ± 9.6 before and after IP, respectively. Before EX2 the ratio was 2.0 ± 2.5 and 2.3 ± 2.3 before and after IP, respectively, and within the first minute of EX2 it peaked at 21.8 ± 0.7 and 16.5 ± 9.1. The estimated in vivo buffer capacity for EX1 and EX2 was 16% (227 ± 67 vs. 195 ± 21 mmol·pH^{-1}·kg dry wt^{-1}) and 19% (251 ± 59 vs. 211 ± 37 vs. mmol·pH^{-1}·kg dry wt^{-1}) higher ($P < 0.05$) after compared with before IP, respectively.

Muscle CP

Muscle CP before EX1 was 95.5 ± 3.9 and 81.7 ± 8.5 mmol/kg dry wt before and after IP, respectively, and decreased by 57.0 ± 6.5 and 54.0 ± 5.6 mmol/kg dry wt during EX1 (Table 3). The decrease in CP during EX2 was also similar before and after IP (40.3 ± 5.4 vs. 40.1 ± 3.7 mmol/kg dry wt), respectively.

Table 3. *Muscle metabolites (mmol·kg dry wt^{-1}) and pH at rest as well as immediately before and after cycling at 20 W followed by EX1 and EX2 in cyclists before (Pre) and after (Post) 7 wk of high-intensity training in combination with a reduced training volume in trained cyclists*

	EX1		EX2	
	Pre	Post	Pre	Post
CP				
Before	95.5 ± 3.9	38.5 ± 3.7	82.2 ± 4.7	41.9 ± 6.4
After	81.7 ± 8.5	27.7 ± 5.5	62.3 ± 4.5*	22.2 ± 3.7*
Lactate				
Before	2.9 ± 0.2	48.8 ± 6.0	38.8 ± 4.3	58.6 ± 8.9
After	14.2 ± 2.7	66.7 ± 7.9†	34.8 ± 4.2	82.5 ± 9.8‡
pH				
Before	7.13 ± 0.04	6.90 ± 0.04	6.95 ± 0.03	6.89 ± 0.05
After	7.05 ± 0.05	6.85 ± 0.05	7.03 ± 0.03	6.81 ± 0.04

Values are means ± SE; $n = 7$ cyclists. CP, creatine phosphate. *After different ($P < 0.05$) from before. †After different ($P < 0.01$) from before. ‡After different ($P < 0.001$) from before.

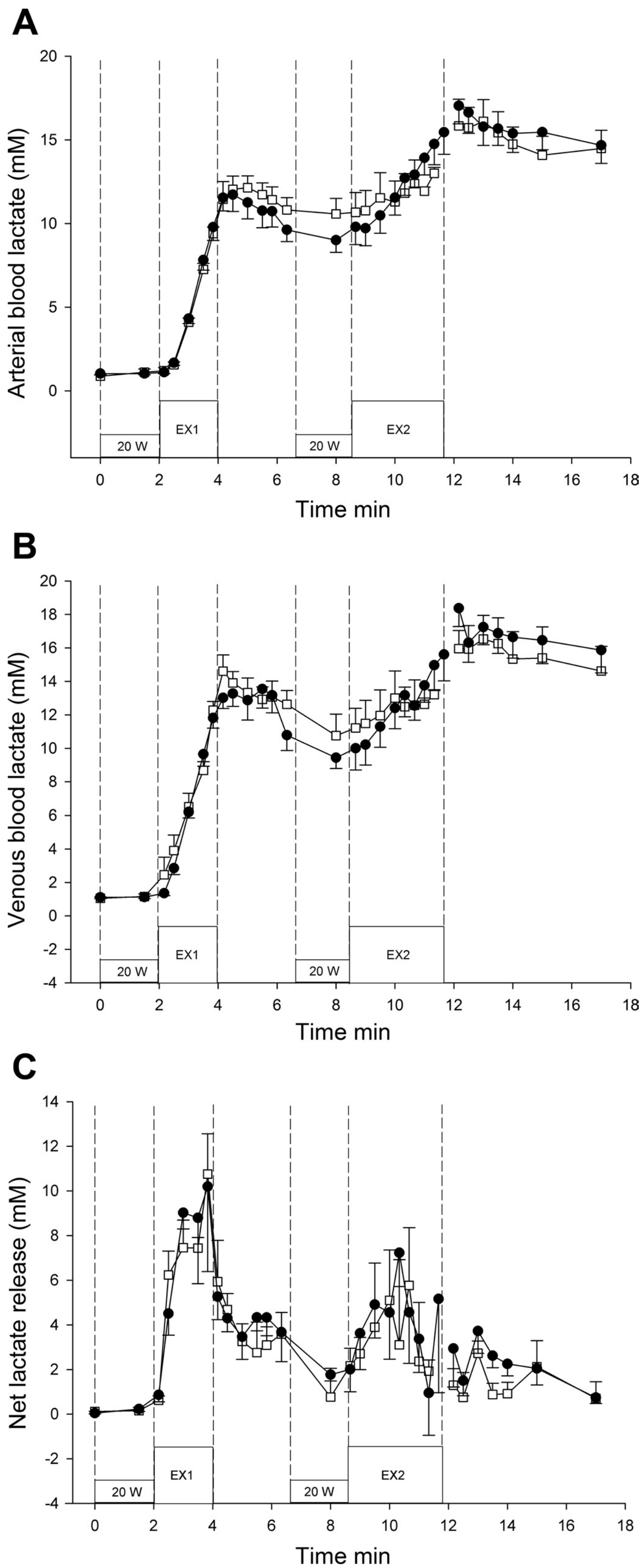

Fig. 8. Arterial (*A*) and venous (*B*) blood lactate and net lactate release (*C*) before, during, and after cycling at 20 W followed by EX1 and EX2 (368 ± 6 W) before (open symbols) and after (filled symbols) 7 wk of high-intensity training in combination with a reduced training volume in trained cyclists (n = 7). Values are means ± SE.

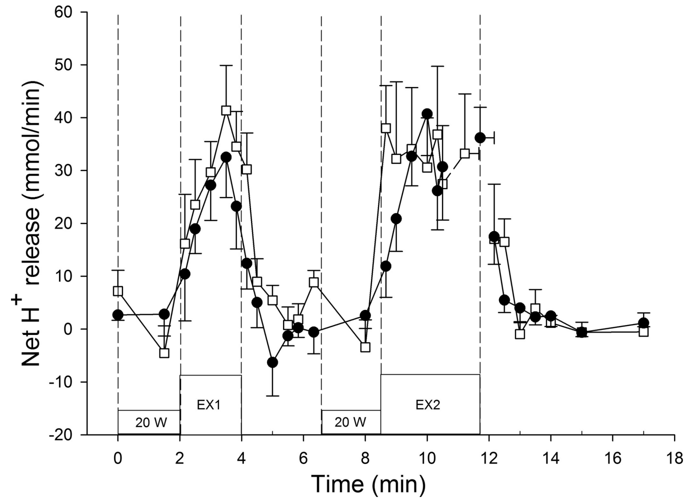

Fig. 9. Net H^+ release before, during, and after cycling at 20 W followed by EX1 and EX2 (368 ± 6 W) before (open symbols) and after (filled symbols) 7 wk of high-intensity training in combination with a reduced training volume in trained cyclists (n = 7). Values are means ± SE.

DISCUSSION

The major findings of the present study were that performance during repeated intense exercise was improved after a period of intensified training in combination with a reduced training volume, and that the femoral venous plasma K^+ concentration in the first 2 min of recovery from EX1 and EX2 was lower after compared with before IP. Furthermore, the intramuscular lactate accumulation and the in vivo buffer capacity during EX1 and EX2 were higher ($P < 0.05$) after compared with before IP.

The lowered femoral venous plasma K^+ concentration observed in recovery from EX1 and EX2 after compared with before IP could be explained by a greater reuptake of K^+ by the previous active muscle cells. This observation is supported by the finding of a lower net K^+ release (4.5 ± 1.0 vs. 1.7 ± 1.6 mmol; Table 1) in recovery from EX2 after compared IP with before IP. The observed lowering of venous plasma K^+ concentrations during and in recovery from exercise may also have been caused by an improved K^+ clearance by the red blood cells (RBC) after IP. However, it is unlikely to be a major factor since the RBC K^+ concentration is unchanged during exercise (19), and alterations in intramuscular K^+ uptake occur unaffected by changes in the RBC K^+ concentration (26). Taken together, the findings suggest an enhanced intramuscular K^+ reuptake during and after intense exercise (EX1 and EX2) after compared with before IP.

The lowered femoral venous K^+ concentration was not associated with alterations in the expression of the Na^+-K^+ pump subunits. In contrast, other studies implementing intense intermittent training have observed elevated levels of the Na^+-K^+ pump subunits (3, 18, 34). In the study by Nielsen et al. (34) untrained subjects had a higher content of the Na^+-K^+ pump subunits α_1, α_2, and β_1 after 8 wk of intense one-legged knee-extensor exercise, and the Na^+-K^+ pump α_1 and α_2 subunits have also been shown to be upregulated in trained runners after 4–9 wk of intensified and reduced training (3, 18). The difference between these studies and the present study may be related to exercise modalities. The biopsy was obtained in *m. vastus lateralis*, which in cycling is activated to a greater extent than during running, and it may be that levels of Na^+-K^+ pump subunits were already high in our subjects before the study in contrast to the studies with runners. Nevertheless, the lack of change in the expression of the Na^+-K^+ pump subunits suggest that other factors have caused the lowered femoral venous K^+ concentration in recovery from exercise. In addition to the Na^+-K^+ pump, several proteins have been shown to be involved in both K^+ release and reuptake in skeletal muscle (1, 9, 41).

The strong inward rectifying K^+ channel (Kir2.1) (5, 22) has been suggested to be of importance for the K^+ reuptake during exercise in human skeletal muscle (23). However, no significant change in the content of Kir2.1 was observed by IP, despite six of eight subjects had higher content and the average content of Kir2.1 was 18% higher after compared with before IP. The content of Kir6.2 was lower (14%; $P < 0.05$) after IP suggesting that Kir6.2 is not of importance for intramuscular K^+ handling during intense exercise in already trained subjects. The Na^+-K^+-$2Cl^-$ exchanger (NKCC1) may have played a role in the altered K^+ handling since in a recent study the expression of NKCC1 was higher (~15%) after 4 wk of speed endurance training in trained runners (18). In the present study we were unable to examine changes in cell surface expression of ion transporting proteins and cannot, as such, exclude the possibility that changes in K^+ handling are in part due to alterations in the cell surface expression of K^+ transporting proteins such as the Na^+-K^+ pump subunits, NKCC1, Kir2.1 and Kir6.2. Nevertheless, it is clear that further investigations elucidating the cause of the lowered femoral venous K^+ concentration after a period of intensified training in combination with a reduced training volume are warranted.

The repeated intense exercise performance was improved by IP. It may have been related to the reuptake of K^+ after EX1 and EX2 appearing to be 2–3 times higher after compared with before IP. In addition, despite the longer exercise time during EX2, muscle interstitial K^+ concentrations tended to be lower at the end of EX2 after IP, which may have delayed the time point for when the muscle fibers were not able to contract due to the membrane potential reaching a critical level of depolarization. Higher anaerobic energy turnover may also have positively affected performance. The intramuscular lactate concentration after EX2 was higher after IP, which also has been demonstrated in studies where untrained subjects carried out intense training (10, 32). In isolated muscle preparations, elevation of extracellular lactate has been shown to prevent loss of excitability in fatiguing muscle (11, 37). Thus, in the present study, the higher intramuscular lactate accumulation observed during EX2 may have delayed fatigue development and could possibly explain the improved repeated intense exercise performance. However, it is not a likely explanation, since the effect is believed to be through a lowering of muscle pH, which was not altered by IP.

In the present study NHE1 was not significantly altered by IP despite a 30% higher content. This finding is in agreement with some (3, 14) but not all (18) studies in trained runners implementing 10–30 s near-maximal intervals for 4–9 wk. Also, no change in H^+ release during intense exercise was observed in the present study. In addition, muscle lactate release during exercise was not altered by IP, despite the observation of a higher intramuscular lactate accumulation at the end of EX1 and EX2. The release of H^+ and lactate (~40 and ~10 mmol/min) during exercise was similar to previous observations during intense one-legged knee-extensor exercise in untrained subjects (20, 39). After IP, the in vivo buffer capacity during EX1 was elevated (~16%; $P < 0.05$) from a prestudy level of 195 ± 21 mmol·pH^{-1}·kg dry wt^{-1}, which is higher than reported in sedentary subjects but similar to levels of subjects active in ball games (164 vs. 194 mmol·pH^{-1}·kg dry wt^{-1}) (40). The higher in vivo buffer capacity may have caused the better performance after IP, but no relationship between muscle buffer capacity and performance during the repeated sprint test was observed, which is in contrast to previous findings (7). However, in the latter study both untrained and trained subjects were included, and no correlation between muscle buffer capacity and repeated sprint performance was observed within the group of trained subjects. It appears that other factors than the in vivo buffer capacity was responsible for the observed increase in mean (4%) and peak (3%) power output and time to exhaustion during EX2 in the present study.

Perspectives and Significance

The present study examined performance and muscular effects in relation to a change from regular endurance training to a reduced amount of training in combination with an intensified training regime. Significant changes in the repeated short-term exercise performance were observed, which was associated with a lowered femoral venous K^+ concentration in recovery from exercise and in vivo buffer capacity without concomitant changes in muscle ion transport proteins and, despite a decrease in content of Kir6.2. This study was the first to investigate training effects on K^+ channels (Kir2.1 and Kir6.2) in humans. Furthermore, it has important practical implications suggesting that performance and ion kinetics can be improved in already trained subjects and, that intramuscular adaptations can occur despite a drastic reduction in training volume. Thus endurance athletes may benefit from replacing some of their endurance training with sessions of high-intensity exercise. This information is of interest not only for elite endurance athletes but also for recreationally active individuals seeking to optimize performance.

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DISCLOSURES

No conflicts of interest, financial or otherwise, are declared by the author(s).

AUTHOR CONTRIBUTIONS

Author contributions: T.P.G., P.M.C., M.T., L.R.N., and J.B. conception and design of research; T.P.G., P.M.C., M.T., L.R.N., and J.B. performed experiments; T.P.G., P.M.C., M.T., L.R.N., and J.B. analyzed data; T.P.G., P.M.C., M.T., L.R.N., and J.B. interpreted results of experiments; T.P.G. and L.R.N. prepared figures; T.P.G. drafted manuscript; T.P.G., P.M.C., M.T., L.R.N., and J.B. approved final version of manuscript; P.M.C. and J.B. edited and revised manuscript.

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